

Fuzzy Logic : Introduction

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What is Fuzzy logic?

- Fuzzy logic is a mathematical language to **express** something.
This means it has grammar, syntax, semantic like a language for communication.
- There are some other mathematical languages also known
 - **Relational algebra** (operations on sets)
 - **Boolean algebra** (operations on Boolean variables)
 - **Predicate logic** (operations on well formed formulae (wff), also called predicate propositions)
- **Fuzzy logic deals with Fuzzy set.**

A brief history of Fuzzy Logic

- First time introduced by **Lotfi Abdelli Zadeh** (1965), University of California, Berkley, USA (1965).



- He is fondly nick-named as **LAZ**

A brief history of Fuzzy logic



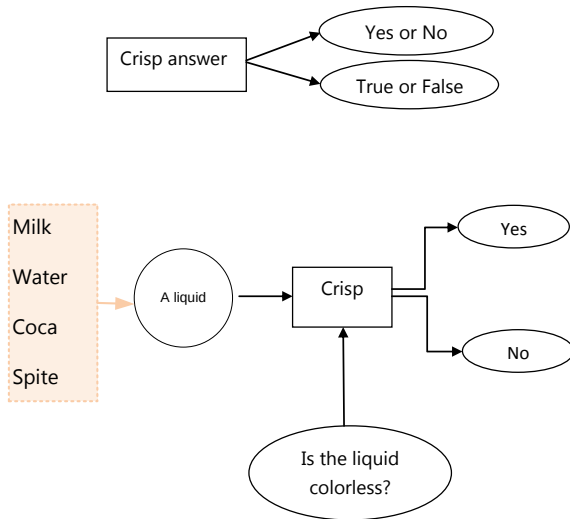
- 1 Dictionary meaning of **fuzzy** is not clear, noisy etc.

Example: Is the picture on this slide is fuzzy?

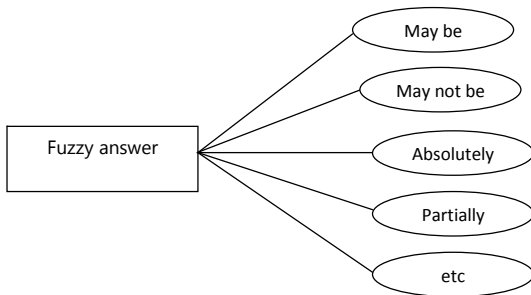
- 2 Antonym of fuzzy is **crisp**

Example: Are the chips crisp?

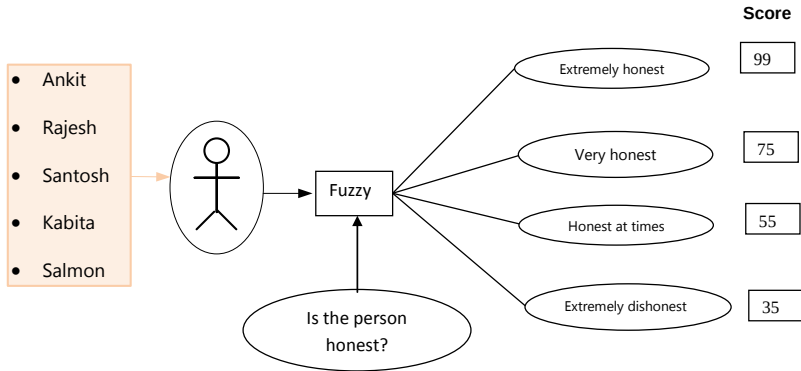
Example : Fuzzy logic vs. Crisp logic



Example : Fuzzy logic vs. Crisp logic



Example : Fuzzy logic vs. Crisp logic

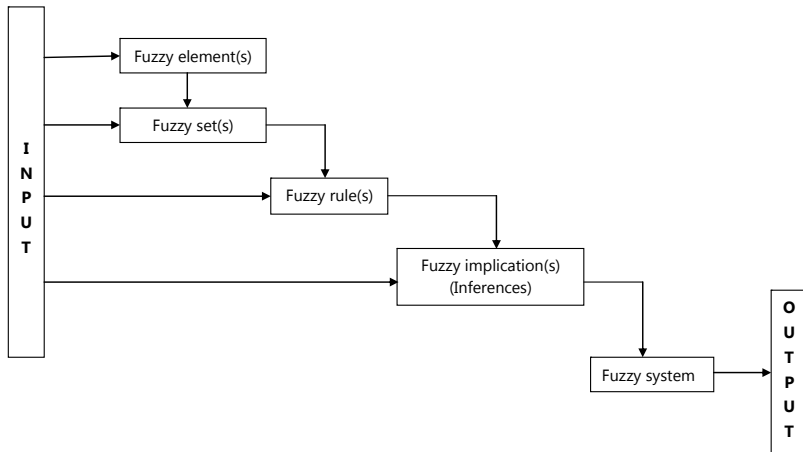


World is fuzzy!



**Our world is better
described with
fuzzily!**

Concept of fuzzy system



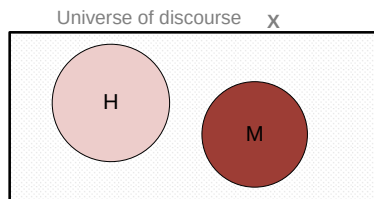
Concept of fuzzy set

To understand the concept of **fuzzy set** it is better, if we first clear our idea of **crisp set**.

X = The entire population of India.

H = All Hindu population = $\{ h_1, h_2, h_3, \dots, h_L \}$

M = All Muslim population = $\{ m_1, m_2, m_3, \dots, m_N \}$



Here, All are the sets of finite numbers of individuals.

Such a set is called **crisp set**.

Example of fuzzy set

Let us discuss about fuzzy set.

X = All students in IT60108.

S = All **Good students**.

$S = \{ (s, g) \mid s \in X \}$ and $g(s)$ is a measurement of goodness of the student s .

Example:

$S = \{ (\text{Rajat}, 0.8), (\text{Kabita}, 0.7), (\text{Salman}, 0.1), (\text{Ankit}, 0.9) \}$ etc.

Fuzzy set vs. Crisp set

Crisp Set	Fuzzy Set
1. $S = \{ s \mid s \in X \}$	1. $F = (s, \mu) \mid s \in X \text{ and } \mu(s) \text{ is the degree of } s.$
2. It is a collection of elements.	2. It is collection of ordered pairs.
3. Inclusion of an element $s \in X$ into S is crisp, that is, has strict boundary yes or no .	3. Inclusion of an element $s \in X$ into F is fuzzy, that is, if present, then with a degree of membership .

Fuzzy set vs. Crisp set

Note: A crisp set is a fuzzy set, but, a fuzzy set is not necessarily a crisp set.

Example:

$$H = \{ (h_1, 1), (h_2, 1), \dots, (h_L, 1) \}$$

$$\text{Person} = \{ (p_1, 1), (p_2, 0), \dots, (p_N, 1) \}$$

In case of a crisp set, the elements are with extreme values of degree of membership namely either 1 or 0.

How to decide the degree of memberships of elements in a fuzzy set?

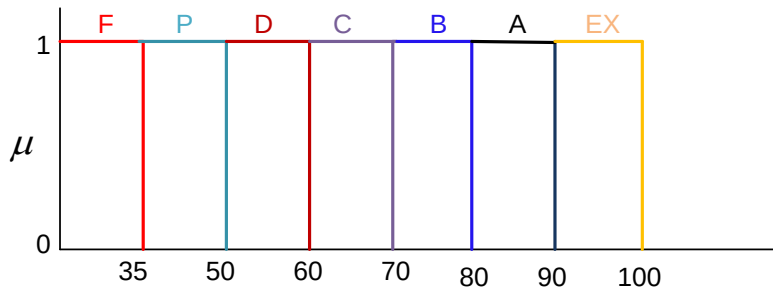
City	Bangalore	Bombay	Hyderabad	Kharagpur	Madras	Delhi
DoM	0.95	0.90	0.80	0.01	0.65	0.75

How the cities of **comfort** can be judged?

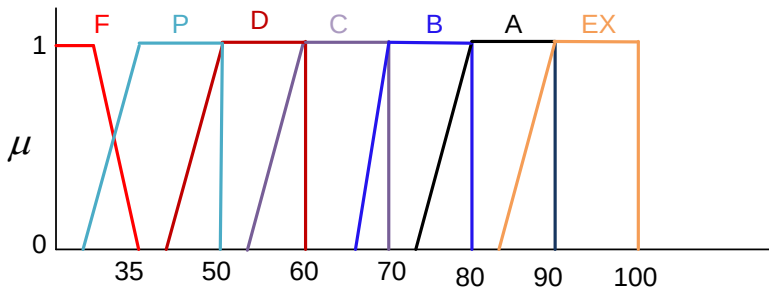
Example: Course evaluation in a crisp way

- ❶ $EX = \text{Marks} \geq 90$
- ❷ $A = 80 \leq \text{Marks} < 90$
- ❸ $B = 70 \leq \text{Marks} < 80$
- ❹ $C = 60 \leq \text{Marks} < 70$
- ❺ $D = 50 \leq \text{Marks} < 60$
- ❻ $P = 35 \leq \text{Marks} < 50$
- ❼ $F = \text{Marks} < 35$

Example: Course evaluation in a crisp way



Example: Course evaluation in a fuzzy way



Few examples of fuzzy set

- High Temperature
- Low Pressure
- Color of Apple
- Sweetness of Orange
- Weight of Mango

Note: Degree of membership values lie in the range $[0...1]$.

Some basic terminologies and notations

Definition 1: Membership function (and Fuzzy set)

If X is a universe of discourse and $x \in X$, then a fuzzy set A in X is defined as a set of ordered pairs, that is

$A = \{(x, \mu_A(x)) | x \in X\}$ where $\mu_A(x)$ is called the **membership function** for the fuzzy set A .

Note:

$\mu_A(x)$ map each element of X onto a membership grade (or membership value) between 0 and 1 (both inclusive).

Question:

How (and who) decides $\mu_A(x)$ for a Fuzzy set A in X ?

Some basic terminologies and notations

Example:

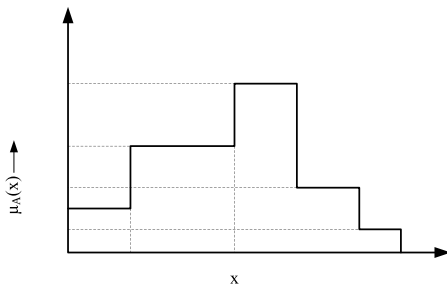
X = All cities in India

A = City of comfort

$A = \{(New\ Delhi, 0.7), (Bangalore, 0.9), (Chennai, 0.8), (Hyderabad, 0.6), (Kolkata, 0.3), (Kharagpur, 0)\}$

Membership function with discrete membership values

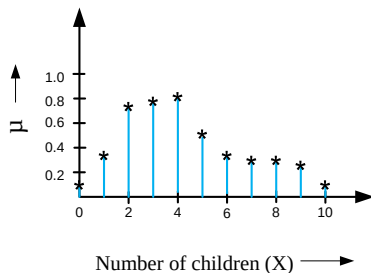
The membership values may be of discrete values.



A fuzzy set with discrete values of μ

Membership function with discrete membership values

Either elements or their membership values (or both) also may be of discrete values.



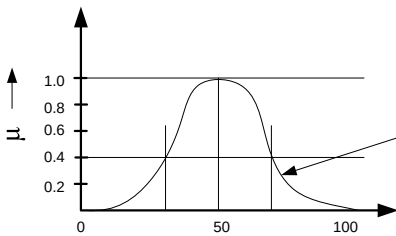
$$A = \{(0, 0.1), (1, 0.35), (2, 0.75), \dots, (10, 0.1)\}$$

Note : X = discrete value

How you measure happiness ??

A = "Happy family"

Membership function with continuous membership values



$$\mu_B(x) = \frac{1}{1 + \left(\frac{x-50}{10}\right)^4}$$

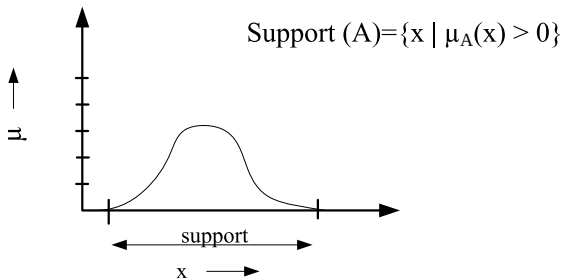
$$B = \{(x, \mu_B(x))\}$$

Note : x = real value
 $= \mathbb{R}^+$

$B = \text{"Middle aged"}$

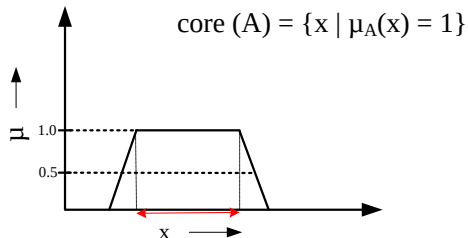
Fuzzy terminologies: Support

Support: The support of a fuzzy set A is the set of all points $x \in X$ such that $\mu_A(x) > 0$



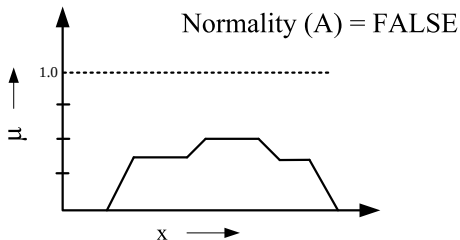
Fuzzy terminologies: Core

Core: The core of a fuzzy set A is the set of all points x in X such that $\mu_A(x) = 1$



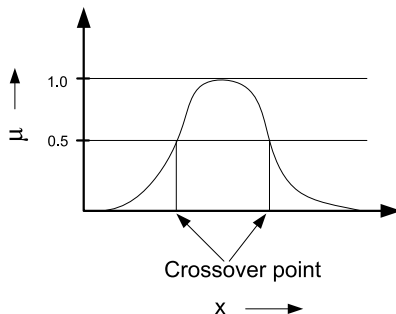
Fuzzy terminologies: Normality

Normality : A fuzzy set A is a normal if its core is non-empty. In other words, we can always find a point $x \in X$ such that $\mu_A(x) = 1$.



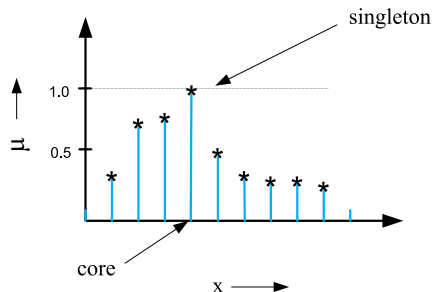
Fuzzy terminologies: Crossover points

Crossover point : A crossover point of a fuzzy set A is a point $x \in X$ at which $\mu_A(x) = 0.5$. That is
 $\text{Crossover}(A) = \{x | \mu_A(x) = 0.5\}$.



Fuzzy terminologies: Fuzzy Singleton

Fuzzy Singleton : A fuzzy set whose support is a single point in X with $\mu_A(x) = 1$ is called a fuzzy singleton. That is $|A| = \{x \mid \mu_A(x) = 1\}$.



Fuzzy terminologies: α -cut and strong α -cut

α -cut and strong α -cut :

The α -cut of a fuzzy set A is a crisp set defined by

$$A_{\alpha} = \{x \mid \mu_A(x) \geq \alpha \}$$

Strong α -cut is defined similarly :

$$A_{\alpha}' = \{x \mid \mu_A(x) > \alpha \}$$

Note : $\text{Support}(A) = A_0'$ and $\text{Core}(A) = A_1$.

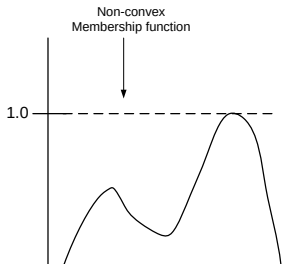
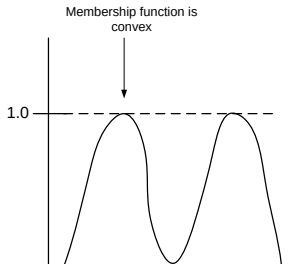
Fuzzy terminologies: Convexity

Convexity : A fuzzy set A is convex if and only if for any x_1 and $x_2 \in X$ and any $\lambda \in [0, 1]$

$$\mu_A(\lambda x_1 + (1 - \lambda)x_2) \geq \min(\mu_A(x_1), \mu_A(x_2))$$

Note :

- A is convex if all its α - level sets are convex.
- Convexity (A_α) $\implies A_\alpha$ is composed of a single line segment only.



Bandwidth :

For a normal and convex fuzzy set, the bandwidth (or width) is defined as the distance the two unique crossover points:

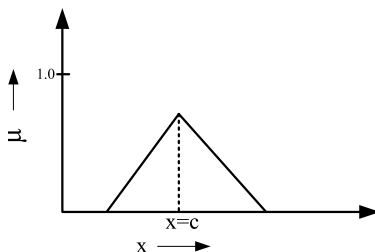
$$\text{Bandwidth}(A) = |x_1 - x_2|$$

where $\mu_A(x_1) = \mu_A(x_2) = 0.5$

Fuzzy terminologies: Symmetry

Symmetry :

A fuzzy set A is symmetric if its membership function around a certain point $x = c$, namely $\mu_A(x + c) = \mu_A(x - c)$ for all $x \in X$.



Fuzzy terminologies: Open and Closed

A fuzzy set A is

Open left

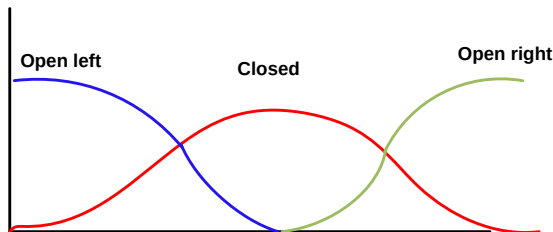
If $\lim_{x \rightarrow -\infty} \mu_A(x) = 1$ and $\lim_{x \rightarrow +\infty} \mu_A(x) = 0$

Open right:

If $\lim_{x \rightarrow -\infty} \mu_A(x) = 0$ and $\lim_{x \rightarrow +\infty} \mu_A(x) = 1$

Closed

If : $\lim_{x \rightarrow -\infty} \mu_A(x) = \lim_{x \rightarrow +\infty} \mu_A(x) = 0$



Fuzzy vs. Probability

Fuzzy : When we say about certainty of a thing

Example: A patient come to the doctor and he has to diagnose so that medicine can be prescribed.

Doctor prescribed a medicine with certainty 60% that the patient is suffering from flue. So, the disease will be cured with certainty of 60% and uncertainty 40%. Here, in stead of flue, other diseases with some other certainties may be.

Probability: When we say about the chance of an event to occur

Example: India will win the T20 tournament with a chance 60% means that out of 100 matches, India own 60 matches.

Prediction vs. Forecasting

The Fuzzy vs. Probability is analogical to Prediction vs. Forecasting

Prediction : When you start guessing about things.

Forecasting : When you take the information from the past job and apply it to new job.

The main difference:

Prediction is based on the best guess from experiences.

Forecasting is based on data you have actually recorded and packed from previous job.

Fuzzy Membership Functions

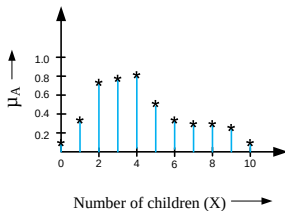
Fuzzy membership functions

A fuzzy set is completely characterized by its membership function (sometimes abbreviated as *MF* and denoted as μ). So, it would be important to learn how a membership function can be expressed (mathematically or otherwise).

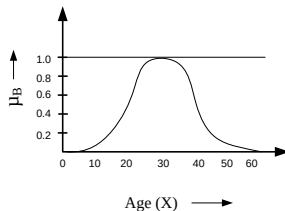
Note: A membership function can be on

- (a) a discrete universe of discourse and
- (b) a continuous universe of discourse.

Example:



A = Fuzzy set of "Happy family"

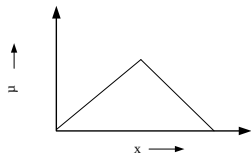


B = "Young age"

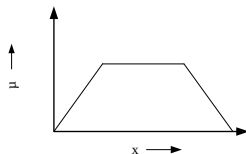
Fuzzy membership functions

So, membership function on a discrete universe of course is trivial. However, a membership function on a continuous universe of discourse needs a special attention.

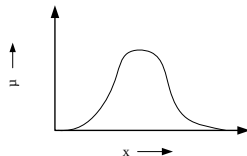
Following figures shows a typical examples of membership functions.



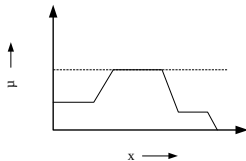
< triangular >



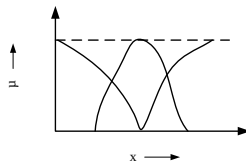
< trapezoidal >



< curve >



< non-uniform >



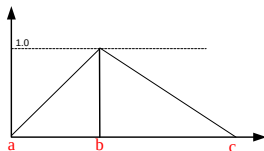
< non-uniform >

Fuzzy MFs : Formulation and parameterization

In the following, we try to parameterize the different MFs on a continuous universe of discourse.

Triangular MFs : A triangular MF is specified by three parameters $\{a, b, c\}$ and can be formulated as follows.

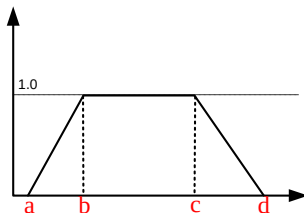
$$\text{triangle}(x; a, b, c) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ \frac{c-x}{c-b} & \text{if } b \leq x \leq c \\ 0 & \text{if } c \leq x \end{cases} \quad (1)$$



Fuzzy MFs: Trapezoidal

A trapezoidal MF is specified by four parameters $\{a, b, c, d\}$ and can be defined as follows:

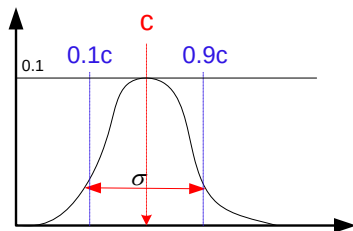
$$\text{trapeziod}(x; a, b, c, d) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } b \leq x \leq c \\ \frac{d-x}{d-c} & \text{if } c \leq x \leq d \\ 0 & \text{if } d \leq x \end{cases} \quad (2)$$



Fuzzy MFs: Gaussian

A Gaussian MF is specified by two parameters $\{c, \sigma\}$ and can be defined as below:

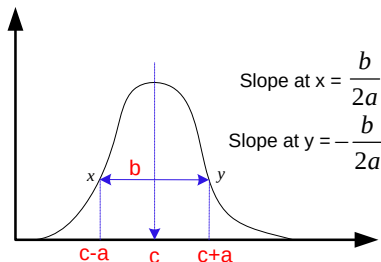
$$\text{gaussian}(x; c, \sigma) = e^{-\frac{1}{2} \left(\frac{x-c}{\sigma} \right)^2}.$$



Fuzzy MFs: Generalized bell

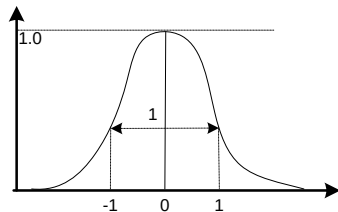
It is also called **Cauchy MF**. A generalized bell MF is specified by three parameters $\{a, b, c\}$ and is defined as:

$$\text{bell}(x; a, b, c) = \frac{1}{1 + \left| \frac{x-c}{a} \right|^{2b}}$$

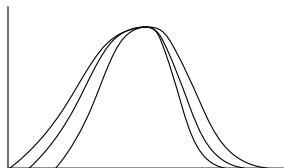


Example: Generalized bell MFs

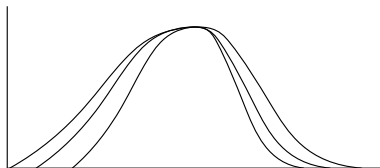
Example: $\mu(x) = \frac{1}{1+x^2}$;
 $a = b = 1$ and $c = 0$;



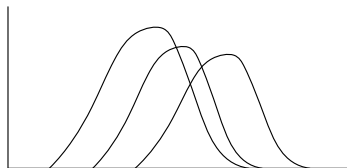
Generalized bell MFs: Different shapes



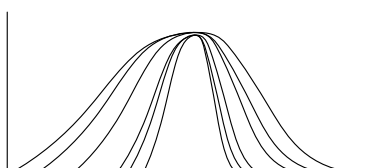
Changing a



Changing b



Changing a

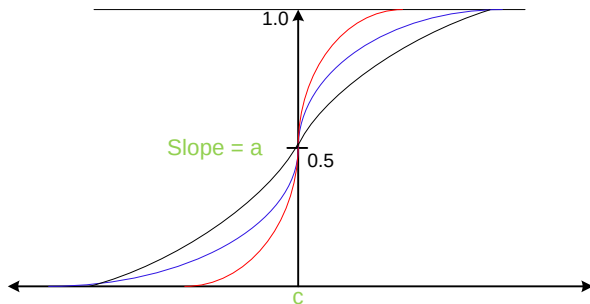


Changing a and b

Fuzzy MFs: Sigmoidal MFs

Parameters: $\{a, c\}$; where c = crossover point and a = slope at c ;

$$\text{Sigmoid}(x;a,c) = \frac{1}{1 + e^{-[\frac{a}{x-c}]}}$$



Fuzzy MFs : Example

Example : Consider the following grading system for a course.

Excellent = Marks ≤ 90

Very good = $75 \leq \text{Marks} \leq 90$

Good = $60 \leq \text{Marks} \leq 75$

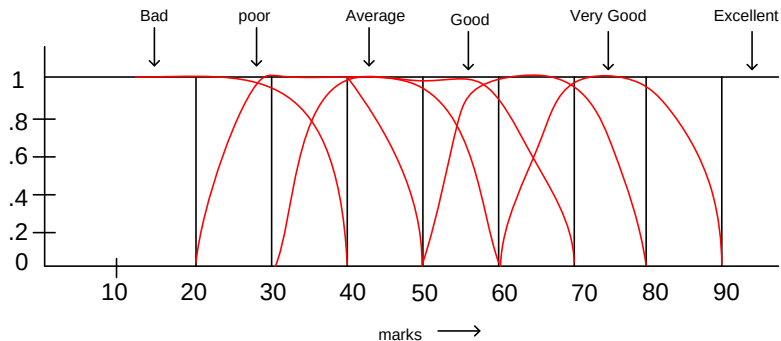
Average = $50 \leq \text{Marks} \leq 60$

Poor = $35 \leq \text{Marks} \leq 50$

Bad = Marks ≤ 35

Grading System

A fuzzy implementation will look like the following.



You can decide a standard fuzzy MF for each of the [fuzzy garde](#).

Operations on Fuzzy Sets

Basic fuzzy set operations: Union

Union ($A \cup B$):

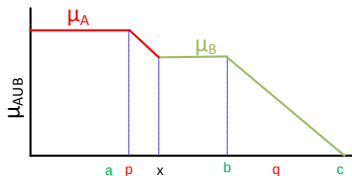
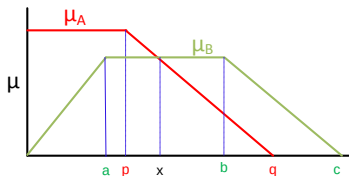
$$\mu_{A \cup B}(x) = \max\{\mu_A(x), \mu_B(x)\}$$

Example:

$A = \{(x_1, 0.5), (x_2, 0.1), (x_3, 0.4)\}$ and

$B = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5)\};$

$C = A \cup B = \{(x_1, 0.5), (x_2, 0.3), (x_3, 0.5)\}$



Basic fuzzy set operations: Intersection

Intersection ($A \cap B$):

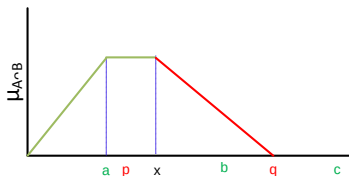
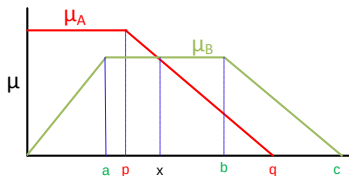
$$\mu_{A \cap B}(x) = \min\{\mu_A(x), \mu_B(x)\}$$

Example:

$A = \{(x_1, 0.5), (x_2, 0.1), (x_3, 0.4)\}$ and

$B = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5)\};$

$C = A \cap B = \{(x_1, 0.2), (x_2, 0.1), (x_3, 0.4)\}$



Basic fuzzy set operations: Complement

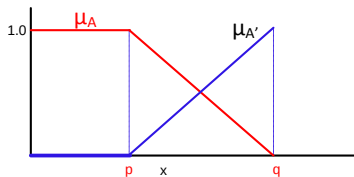
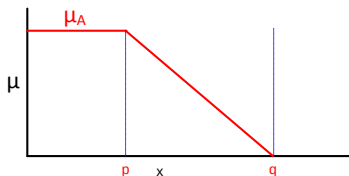
Complement (A^C):

$$\mu_{A^C}(x) = 1 - \mu_A(x)$$

Example:

$$A = \{(x_1, 0.5), (x_2, 0.1), (x_3, 0.4)\}$$

$$C = A^C = \{(x_1, 0.5), (x_2, 0.9), (x_3, 0.6)\}$$



Basic fuzzy set operations: Products

Algebraic product or Vector product ($A \bullet B$):

$$\mu_{A \bullet B}(X) = \mu_A(X) \bullet \mu_B(X)$$

Scalar product ($\alpha \times A$):

$$\mu_{\alpha A}(X) = \alpha \cdot \mu_A(X)$$

Basic fuzzy set operations: Sum and Difference

Sum ($A + B$):

$$\mu_{A+B}(x) = \mu_A(x) + \mu_B(x) - \mu_A(x) \cdot \mu_B(x)$$

Difference ($A - B = A \cap B^C$):

$$\mu_{A-B}(x) = \mu_{A \cap B^C}(x)$$

Disjunctive sum: $A \oplus B = (A^C \cap B) \cup (A \cap B^C)$

Bounded Sum: $|A(x) \oplus B(x)|$

$$\mu_{|A(x) \oplus B(x)|} = \min\{1, \mu_A(x) + \mu_B(x)\}$$

Bounded Difference: $|A(x) \ominus B(x)|$

$$\mu_{|A(x) \ominus B(x)|} = \max\{0, \mu_A(x) + \mu_B(x) - 1\}$$

Basic fuzzy set operations: Equality and Power

Equality ($A = B$):

$$\mu_A(X) = \mu_B(X)$$

Power of a fuzzy set A^α :

$$\mu_{A^\alpha}(X) = \{\mu_A(X)\}^\alpha$$

- If $\alpha < 1$, then it is called *dilation*
- If $\alpha > 1$, then it is called *concentration*

Basic fuzzy set operations: Cartesian product

Cartesian Product ($A \times B$):

$$\mu_{A \times B}(x, y) = \min\{\mu_A(x), \mu_B(y)\}$$

Example 3:

$$A(x) = \{(x_1, 0.2), (x_2, 0.3), (x_3, 0.5), (x_4, 0.6)\}$$

$$B(y) = \{(y_1, 0.8), (y_2, 0.6), (y_3, 0.3)\}$$

$$A \times B = \min\{\mu_A(x), \mu_B(y)\} =$$

	y_1	y_2	y_3
x_1	0.2	0.2	0.2
x_2	0.3	0.3	0.3
x_3	0.5	0.5	0.3
x_4	0.6	0.6	0.3

Properties of fuzzy sets

Commutativity :

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Associativity :

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Distributivity :

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Properties of fuzzy sets

Idempotence :

$$A \cup A = A$$

$$A \cap A = \emptyset$$

$$A \cup \emptyset = A$$

$$A \cap \emptyset = \emptyset$$

Transitivity :

If $A \subseteq B, B \subseteq C$ then $A \subseteq C$

Involution :

$$(A^c)^c = A$$

De Morgan's law :

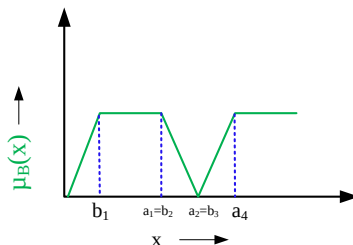
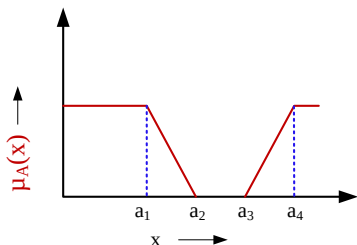
$$(A \cap B)^c = A^c \cup B^c$$

$$(A \cup B)^c = A^c \cap B^c$$

Few Illustrations on Fuzzy Sets

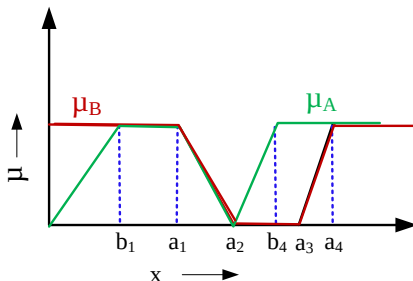
Example 1: Fuzzy Set Operations

Let A and B are two fuzzy sets defined over a universe of discourse X with membership functions $\mu_A(x)$ and $\mu_B(x)$, respectively. Two MFs $\mu_A(x)$ and $\mu_B(x)$ are shown graphically.



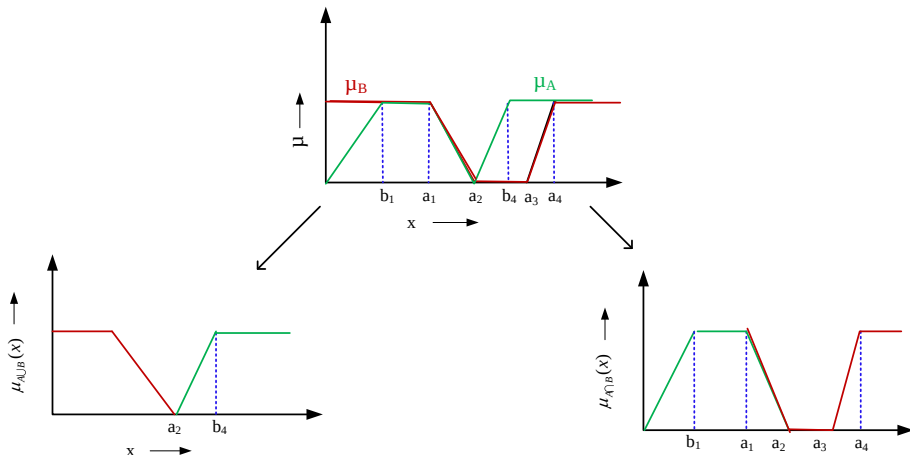
Example 1: Plotting two sets on the same graph

Let's plot the two membership functions on the same graph



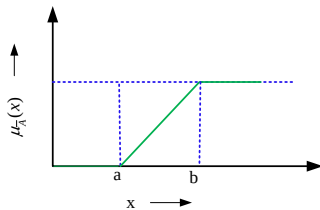
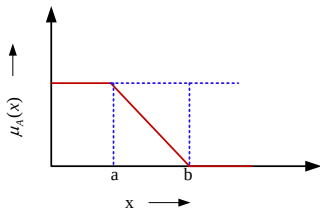
Example 1: Union and Intersection

The plots of union $A \cup B$ and intersection $A \cap B$ are shown in the following.



Example 1: Intersection

The plots of union $\mu_{\bar{A}}(x)$ of the fuzzy set A is shown in the following.



Fuzzy set operations: Practice

Consider the following two fuzzy sets A and B defined over a universe of discourse $[0,5]$ of real numbers with their membership functions

$$\mu_A(x) = \frac{x}{1+x} \text{ and } \mu_B(x) = 2^{-x}$$

Determine the membership functions of the following and draw them graphically.

i. $\overline{A}, \overline{B}$

ii. $A \cup B$

iii. $A \cap B$

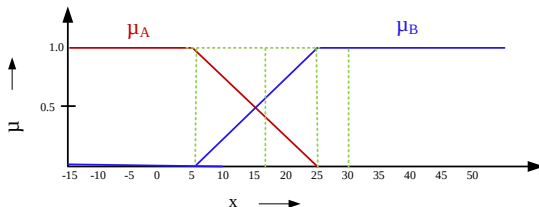
iv. $(A \cup B)^c$ [Hint: Use De' Morgan law]

Example 2: A real-life example

Two fuzzy sets A and B with membership functions $\mu_A(x)$ and $\mu_B(x)$, respectively defined as below.

$A = \text{Cold climate}$ with $\mu_A(x)$ as the MF.

$B = \text{Hot climate}$ with $\mu_B(x)$ as the M.F.



Here, X being the universe of discourse representing entire range of temperatures.

Example 2: A real-life example

What are the fuzzy sets representing the following?

- 1 **Not cold climate**
- 2 **Not hold climate**
- 3 **Extreme climate**
- 4 **Pleasant climate**

Note: Note that "Not cold climate" $\neq$ "Hot climate" and vice-versa.

Example 2 : A real-life example

Answer would be the following.

❶ **Not cold climate**

$\bar{A}$ with $1 - \mu_A(x)$ as the MF.

❷ **Not hot climate**

$\bar{B}$ with $1 - \mu_B(x)$ as the MF.

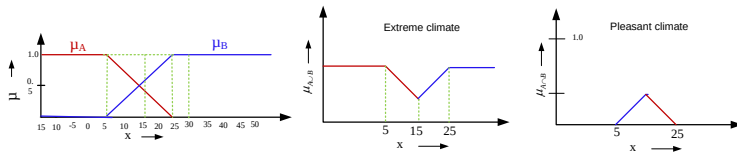
❸ **Extreme climate**

$A \cup B$ with $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$ as the MF.

❹ **Pleasant climate**

$A \cap B$ with $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$ as the MF.

The plot of the MFs of $A \cup B$ and $A \cap B$ are shown in the following.



Few More on Membership Functions

Generation of MFs

Given a membership function of a fuzzy set representing a **linguistic hedge**, we can derive many more MFs representing several other linguistic hedges using the concept of **Concentration** and **Dilation**.

- **Concentration:**

$$A^k = [\mu_A(x)]^k ; k > 1$$

- **Dilation:**

$$A^k = [\mu_A(x)]^k ; k < 1$$

Example : Age = { Young, Middle-aged, Old }

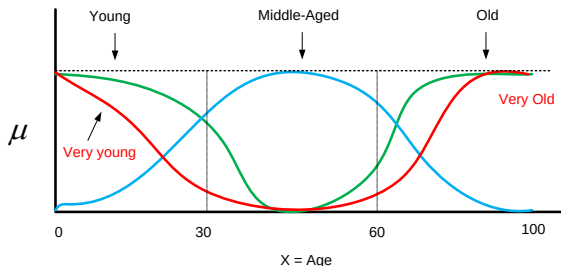
Thus, corresponding to Young, we have : Not young, Very young, Not very young and so on.

Similarly, with Old we can have : old, very old, very very old, extremely old etc.

Thus, **Extremely old** = $((old)^2)^2$ and so on

Or, **More or less old** = $A^{0.5} = (old)^{0.5}$

Linguistic variables and values



$$\mu_{\text{young}}(x) = \text{bell}(x, 20, 2, 0) = \frac{1}{1 + (\frac{x}{20})^4}$$

$$\mu_{\text{old}}(x) = \text{bell}(x, 30, 3, 100) = \frac{1}{1 + (\frac{x-100}{30})^6}$$

$$\mu_{\text{middle-aged}} = \text{bell}(x, 30, 60, 50)$$

$$\text{Not young} = \overline{\mu_{\text{young}}(x)} = 1 - \mu_{\text{young}}(x)$$

$$\text{Young but not too young} = \mu_{\text{young}}(x) \cap \overline{\mu_{\text{young}}(x)}$$

Any questions??